Weisi Chen

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Brief Biography

I hold a Bachelor of Pharmacy from the University of Sydney and a Master of Health Data Science from UNSW. I have over three years of research experience in epidemiology, population health and applied science.

I have experience working with large, linked administrative health datasets. I am passionate in statistical modelling, data visualisation, and data storytelling to translate complex data into meaningful and actionable insights.

Technical Skills & Expertise

SAS · R · R Shiny · GitHub
Big Data Management · Reproducible Research · Data Visualization · Statistical Modelling · Data Storytelling

Education

Sept 2021 - May 2023 **The University of New South Wales** **M.S. Health Data Science**
WAM: 88% (Dissertation: 91%)
Dissertation: Changes in atherosclerotic cardiovascular disease risk over time among people living with HIV

Mar 2015 - Mar 2019 **The University of Sydney** **B.S. Pharmacy**

Employment History

Research-focused Positions

Mar 2026 - Present **Research Officer, NeuRA, UNSW**

- Led one systematic review and contributed to some systematic reviews investigating sex differences in dementia risk factors.
- Lead analyst for a population-based longitudinal cohort study using data from the PATH Through Life Study.
- Contributed to the international CARE project conducting sex-specific prevalence analyses of dementia risk factors to inform population-attributable risk estimations.

Mar 2023 - Dec 2023; April 2025 - Mar 2026 **Data Scientist, Sydney Pharmacy School, USYD**

- Lead analyst and first author of a nationwide cross-sectional data linkage study examining sex-specific multimorbidity patterns among Australians aged 65 years and older.
- Lead analyst for a Multi-Agency Data Integration Project (MADIP) examining high-risk prescribing patterns among older Australians, including those living with dementia, using linked PBS and Census data.
- Lead analyst in the development of a national dementia cohort integrating linked hospital admissions, emergency department presentations, PBS, MBS, aged care, and mortality data.

Jul 2023 – Mar 2025 **Data Scientist, School of Population Health, UNSW**

- Lead analyst for the Aboriginal Community Controlled Ear Health Support System (ACCESS) project, examining temporal trends and patterns in ear-related procedures among Aboriginal children.
- Lead analyst for the evaluation of the Sustaining NSW Families program, assessing program outcomes using population health data.
- Contributed to the development of a large-scale linked data platform integrating multiple administrative datasets for the NSW Child E-Cohort Project.

Clinical Experience

Nov 2018 – Jul 2025 **Registered Pharmacist, Community Pharmacy**

Research Interests

My research interests involve in pharmacoepidemiology and ageing, with a particular focus on developing and applying advanced statistical and analytical methods to large-scale linked health administrative data.

Publications

Google Scholar Profile

Peer-reviewed Journal Articles

1. **Chen, W.**, Lu, C. Y., Hilmer, S., Gibson, A., Tan., E. C. K (2026) "Multimorbidity Clusters Among People Aged 65 Years and Over in Australia: A Nationwide Cross-Sectional Data Linkage Study" *Medical journal of Australia*. 224(7):e70240.. DOI: 10.5694/mja2.70240
2. **Chen, W.**, Petoumenos, K., Somia, A., Edmiston, N., Chaiwarith, R., Woolley, I., Ross, J., Pujari, S., & Boettiger, D. C (2024) "Changes in atherosclerotic cardiovascular disease risk over time among people living with HIV" *Journal of Antimicrobial Chemotherapy*. 79(4), 897–902. DOI: 10.1093/jac/dkae049
3. Lau, E. C. Y., **Chen, W.**, Hilmer, S. N., Jeon, Y. H., Yiu C. H, Lu, C. Y., & Tan, E. C. K. (2026) "Clusters of Health Service and Psychotropic Use and Mortality Risk in People With Dementia: An Australian Cohort Study" *Journal of the American Medical Directors Association*. DOI: 10.1016/j.jamda.2026.106502
4. Lau, E. C. Y., Shim. G., Yiu C. H, Hilmer, S. N., Jeon, Y. H., **Chen, W.**, Lu, C. Y., & Tan, E. C. K. (2026) "Antipsychotic Drug Interactions and Short-Term Mortality Risk in People With Dementia: A National Cohort Study" *Clinical Pharmacology & Therapeutics*. 120(4):1081–1091. DOI: 10.1002/cpt.70407
5. Lau, E. C. Y., **Chen, W.**, Hilmer, S. N., Jeon, Y.-H., Lu, C. Y., Fujita, K., Wang & Tan, E. C. K. (2025) "High-risk geriatric prescribing in older people with and without dementia in Australia: a nationwide cross-sectional study" *Journal of the American Medical Directors Association*. 27(2):106006. DOI: 10.1016/j.jamda.2025.106006
6. Le, H. T., Lau, E. C. Y., **Chen, W.**, Lu, C. Y., Nguyen, T. A., Low, L.-F., Hilmer, S. N., Jeon, Y.-H., & Tan, E. C. K. (2025) "Prevalence and Risk Factors for Psychotropic Medication Use in Older Adults in Australia: A Nationwide Data Linkage Study" *Drugs & Aging*. 42(7):755–769. DOI: 10.1007/s40266-025-01220-6

7. Huynh, A. L. H., Wang, Y., Ma, L., Low, Y. C. L., **Chen, W.**, Fowler, C., Tan, E. C. K., Masters, C. L., Jin, L., Pan, Y., & the AIBL Research Group. (2024) "A comparison of an Australian observational longitudinal Alzheimer's Disease cohort to Community-Based Australian data." *Journal of Alzheimer S Disease*. 1–13. DOI: 10.3233/JAD-240241
8. Lau, E. C. Y., **Chen, W.**, Lu, C. Y., Hilmer, S. N., Jeon, Y.-H., & Tan, E. C. K. (2024) "Antidementia and psychotropic drug use in older people with dementia in Australia: a national data linkage study" *Journal of the American Medical Directors Association*. 25 (11), 105237. DOI: 10.1016/j.jamda.2024.105237